

How Ballistic Missiles Work

A quick overview of principles, flight phases, and classification

What Makes It “Ballistic”

Unlike a cruise missile, which is powered and actively guided for its entire flight, a ballistic missile is only powered during the early part of its journey. Once the engines cut off, it follows a curved, gravity-driven path — much like a thrown ball, but at far greater speed and altitude. This unpowered arc is what gives the missile its name.

The Three Flight Phases

1. Boost Phase Rocket engines (solid or liquid fuel, often in multiple stages that burn and separate) accelerate the missile upward and outward. This is the only powered portion of the flight, typically lasting just a few minutes.	2. Midcourse Phase Engines shut off and the missile coasts unpowered through the upper atmosphere or space along its ballistic arc, guided by pre-set inertial navigation, sometimes refined with satellite positioning data.	3. Terminal Phase The reentry vehicle carrying the warhead separates and dives back into the atmosphere toward the target at extreme speed, shielded by a heat-resistant covering.
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Core Components

- **Propulsion:** Multi-stage propulsion — rocket stages that burn in sequence and drop away once spent, reducing weight and increasing range.
- **Guidance:** Guidance and navigation — primarily inertial systems using gyroscopes and accelerometers to track position from the moment of launch.
- **Payload:** Payload — a conventional or nuclear warhead housed inside the reentry vehicle.

Flight Path Diagram



Simplified trajectory showing the three flight phases and stage separation.